

Module 1: Work Order Management

Tab 11: GIS Update

1. Overview

This tab provides a streamlined, map-centric interface for updating SEC's Geographic Information System (GIS). The workflow is designed to be highly visual, using the specific set of geo-tagged photos taken during the **Electrical** phase as the primary source of truth for the location and type of newly installed assets.

2. Workflow

- 1 **Map Initialization:** When the user opens this tab, a map interface is loaded, centered on the project's location.
- 2 **Photo Plotting:** The system automatically retrieves all photos from the **Electrical** tab that were taken as part of the "Images Required for Geographic Information Systems (GIS)" checklist. It then plots an icon on the map for each distinct asset or location captured in these photos, using the GPS coordinates embedded in each photo's metadata.
- 3 **Interactive Update:** The GIS operator clicks on an icon on the map.
- 4 **Photo Display:** A pop-up modal or side panel appears, displaying the photo(s) associated with that specific point.
- 5 **Manual GIS Entry:** The operator uses the visual information from the photo (e.g., asset type, orientation, visible serial numbers) to manually update the external GIS system.
- 6 **Mark as Complete:** After updating the GIS system for that point, the operator marks the point as complete within our application (e.g., by clicking a "Mark as Updated" button). The icon on the map then changes color (e.g., from red to green) to provide a clear visual indication of progress.

3. Interface

- **Main View:** A full-screen interactive map (e.g., using Leaflet.js, OpenLayers, or a similar library).
- **Map Icons:** Each asset photo location will be represented by an icon. The icon should ideally represent the type of asset (e.g., a small transformer icon, a cabinet icon).
- **Photo Viewer:** The pop-up modal will display the high-resolution image, allowing the operator to zoom and pan for detail. The photo's metadata (Date, Time, Coordinates, Compass Direction) will be displayed clearly below the image.
- **Checklist/Progress Tracker:** A simple list or progress bar could be displayed alongside the map, showing the total number of points and how many have been completed (e.g., "12 of 15 points updated").

4. Completion Logic

- This tab is marked as complete once all photo icons on the map have been marked as "Updated."
- Upon completion, the **GIS** milestone on the main Work Order dashboard is marked as complete.